

PROGRAM 1: MUTUAL EXCLUSION USING SEMAPHORE

C PROGRAM

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <semaphore.h>

#include <sys/mman.h>

#include <sys/wait.h>

int main() {

    sem_t *sem;

    sem = mmap(NULL, sizeof(sem_t),

               PROT_READ | PROT_WRITE,

               MAP_SHARED | MAP_ANONYMOUS,

               -1, 0);

    sem_init(sem, 1, 1);

    if (fork() == 0) {

        sem_wait(sem);

        printf("Child Process Entering Critical Section\n");

        sleep(3);

        printf("Child Process Leaving Critical Section\n");

        sem_post(sem);

        exit(0);

    }

    sem_wait(sem);

    printf("Parent Process Entering Critical Section\n");

    sleep(3);

    printf("Parent Process Leaving Critical Section\n");

    sem_post(sem);
```

```
wait(NULL);  
sem_destroy(sem);  
return 0;  
}
```

SHELL SCRIPT

```
GNU nano 8.7.1  
#!/bin/bash  
  
LOCKFILE="/tmp/mylock"  
  
while [ -f "$LOCKFILE" ]  
do  
    sleep 1  
done  
  
touch "$LOCKFILE"  
  
echo "Entering Critical Section"  
  
sleep 5  
  
echo "Leaving Critical Section"  
  
rm -f "$LOCKFILE"
```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~  
$ nano semaphore.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ chmod +x semaphore.sh  
  
(ganesh@LAPTOP-R3RDH9NG)~  
$ bash semaphore.sh  
Entering Critical Section  
2  
Leaving Critical Section
```